

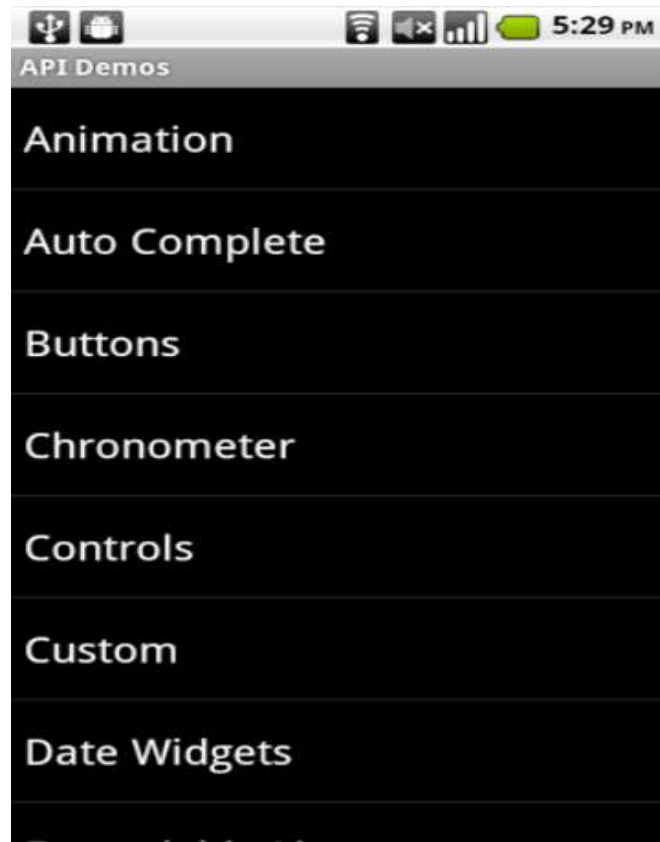
RecyclerView

Contents

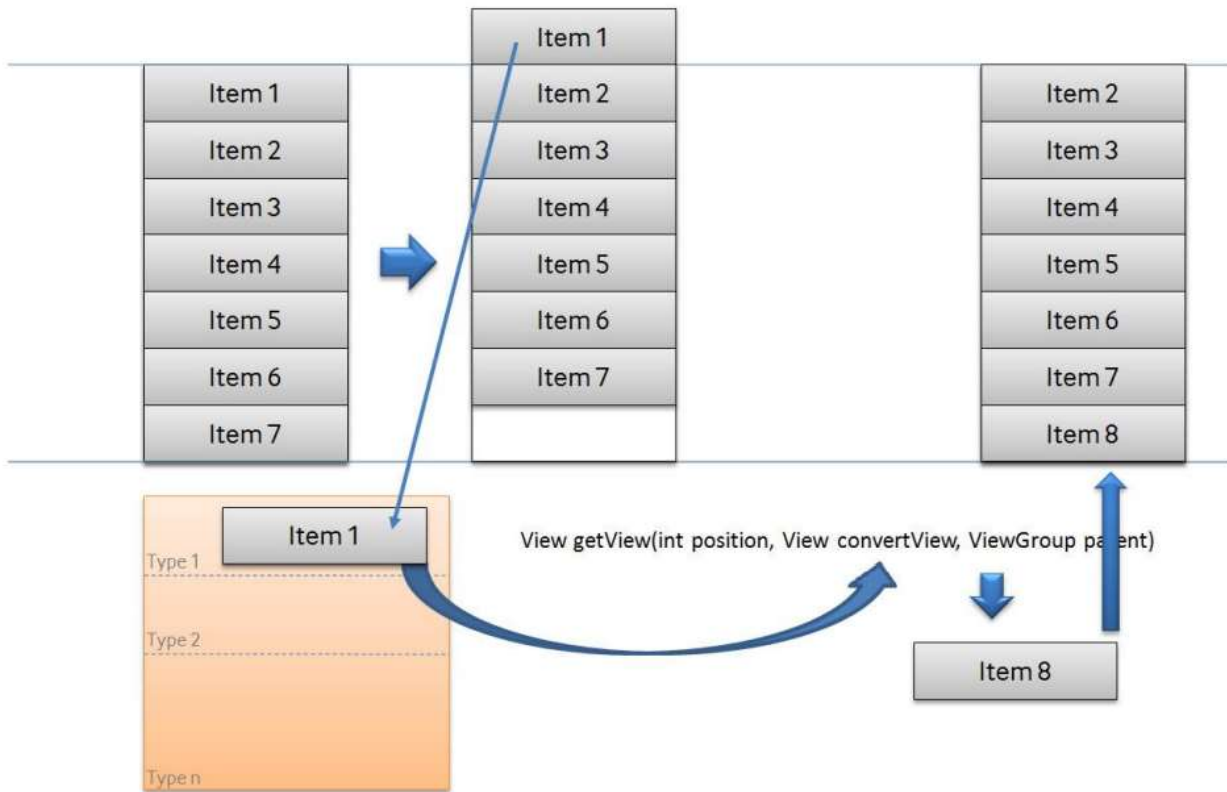
- **What is RecyclerView?**
- **Implementing a RecyclerView**

What is RecyclerView?

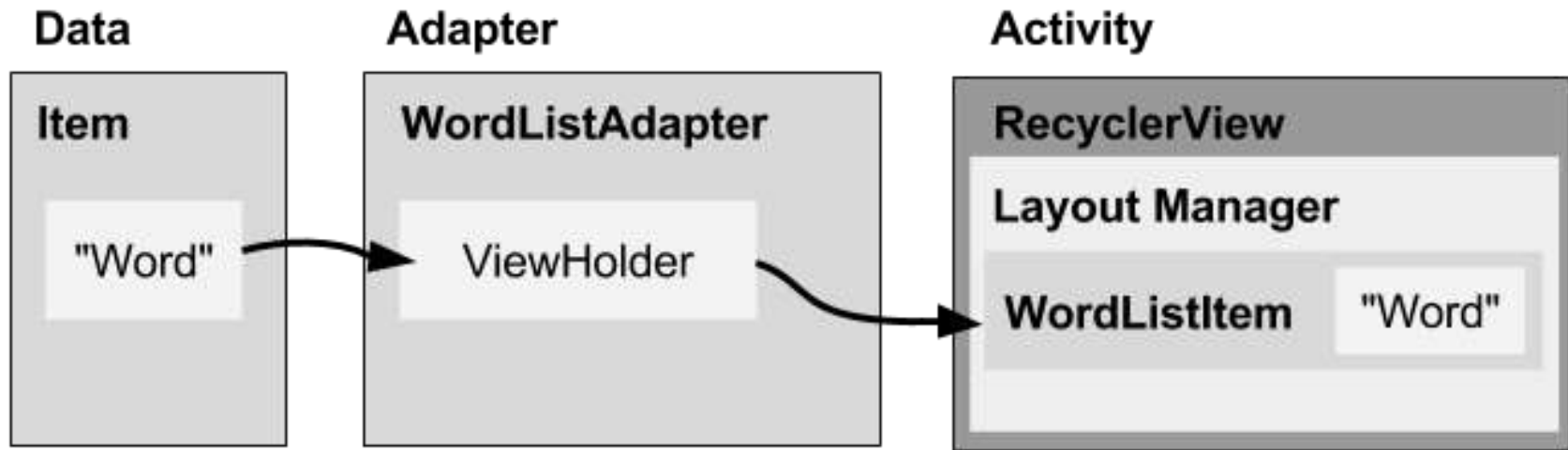
- RecyclerView is a view capable of displaying scrollable list of items
- It only creates views (widget) if needed
- It recycles views, if a row is not displayed anymore, it will be recycled and only its content will change.
- Efficient
 - Uses and reuses limited number of View elements
 - Updates changing data fast



View recycling



RecyclerView components



RecyclerView components

- Data
- RecyclerView scrolling list for list items—[RecyclerView](#)
- Layout for one item of data—XML file
- Layout manager handles the organization of UI components in a View—[Recyclerview.LayoutManager](#)
- Adapter connects data to the RecyclerView—[RecyclerView.Adapter](#)
- ViewHolder has view information for displaying one item—[RecyclerView.ViewHolder](#)

Layout manager

- Each **ViewGroup** has a layout manager
- Use to position **View** items inside a [RecyclerView](#)
- Reuses **View** items that are no longer visible to the user
- Built-in layout managers
 - [LinearLayoutManager](#)
 - [GridLayoutManager](#)
 - [StaggeredGridLayoutManager](#)
- Extend [RecyclerView.LayoutManager](#)

Adapter

- **Helps incompatible interfaces work together**
 - Example: Takes data from database [Cursor](#) and prepares strings to put into a View
- **Intermediary between data and View**
- **Manages creating, updating, adding, deleting View items as underlying data changes**
- [RecyclerView.Adapter](#)

ViewHolder

- Used by the adapter to prepare one View with data for one list item
- Layout specified in an XML resource file
- Can have clickable elements
- Is placed by the layout manager
- [RecyclerView.ViewHolder](#)

Implementing RecyclerView

1. Add RecyclerView dependency to build.gradle if needed
2. Add RecyclerView to layout
3. Create XML layout for item
4. Extend RecyclerView.Adapter
5. Extend RecyclerView.ViewHolder
6. In Activity onCreate(), create RecyclerView with adapter and layout manager

1. Add dependency to app/build.gradle

Add RecyclerView dependency to build.gradle if needed:

```
dependencies {  
    ...  
    compile 'com.android.support:recyclerview-v7:26.1.0'  
    ...  
}
```

2. Add RecyclerView to XML Layout

```
<android.support.v7.widget.RecyclerView  
    android:id="@+id/recyclerview"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent">  
</android.support.v7.widget.RecyclerView>
```

3. Create layout for 1 list item

```
<LinearLayout ...>
```

```
<TextView
```

```
    android:id="@+id/word"
```

```
    style="@style/word_title" />
```

```
</LinearLayout>
```



4. Implement the adapter

```
public class WordListAdapter
    extends RecyclerView.Adapter<WordListAdapter.WordViewHolder> {

    public WordListAdapter(Context context,
                           LinkedList<String> wordList) {
        mInflater = LayoutInflater.from(context);
        this.mWordList = wordList;
    }
}
```

Adapter has 3 required methods

- `onCreateViewHolder()`
- `inBindViewHolder()`
- `getItemCount()`

onCreateViewHolder()

@Override

```
public WordViewHolder onCreateViewHolder(  
    ViewGroup parent, int viewType) {  
    // Create view from layout  
    View mItemView = mInflater.inflate(  
        R.layout.wordlist_item, parent, false);  
    return new WordViewHolder(mItemView, this);  
}
```


onBindViewHolder()

@Override

```
public void onBindViewHolder(  
    WordViewHolder holder, int position) {  
    // Retrieve the data for that position  
    String mCurrent = mWordList.get(position);  
    // Add the data to the view  
    holder.wordItemView.setText(mCurrent);  
}
```

getItemCount()

@Override

```
public int getItemCount() {  
    // Return the number of data items to display  
    return mWordList.size();  
}
```

5. Create the view holder in adapter class

```
class WordViewHolder extends RecyclerView.ViewHolder { //.. }
```

If you want to handle mouse clicks:

```
class WordViewHolder extends RecyclerView.ViewHolder  
    implements View.OnClickListener { //.. }
```

View holder constructor

```
public WordViewHolder(View itemView, WordListAdapter adapter) {  
    super(itemView);  
    // Get the layout  
    wordItemView = itemView.findViewById(R.id.word);  
    // Associate with this adapter  
    this.mAdapter = adapter;  
    // Add click listener, if desired  
    itemView.setOnClickListener(this);  
}  
// Implement onClick() if desired
```


Classwork

- Use RecyclerView to show list of products in HomeActivity
- Click an Item, show DetailActivity



References

- [Using the RecyclerView | CodePath Android Cliffnotes](#)
- <https://google-developer-training.github.io/android-developer-fundamentals-course-concepts-v2/>